

Xylid Tablet 250 mg

Cefuroxime 250mg

DESCRIPTION

White, oblong tablet with Pharmaniaga icon on one side and plain on the other.

COMPOSITION

Each tablet contains Cefuroxime 250 mg (as Cefuroxime axetil).

PHARMACODYNAMICS

The prevalence of acquired resistance is geographically and time dependent and for selected species may be very high. Local information on resistance is desirable, particularly when treating severe infections.

In vitro susceptibility of micro-organisms to Cefuroxime

Where clinical efficacy of cefuroxime axetil has been demonstrated in published clinical trials is indicated with an asterisk (*).

Commonly Susceptible Species

Gram-Positive Aerobes: *Staphylococcus aureus* (methicillin susceptible)*, *Coagulase negative staphylococcus* (methicillin susceptible), *Streptococcus pyogenes**, *Beta-hemolytic streptococci*.

Gram-Negative Aerobes:

*Haemophilus influenzae** including ampicillin resistant strains, *Haemophilus parainfluenzae**, *Moraxella catarrhalis**, *Neisseria gonorrhoea** including penicillinase and non-penicillinase producing strains.

Gram-Positive Anaerobes: *Peptostreptococcus* spp., *Propionibacterium* spp.

Spirochetes: *Borrelia burgdorferi**.

Organisms for which acquired resistance may be a problem are:

Gram-Positive Aerobes: *Streptococcus pneumoniae**.

Gram-Negative Aerobes: *Citrobacter* spp. not including *C. freundii*, *Enterobacter* spp. not including *E. aerogenes* and *E. cloacae*, *Escherichia coli**, *Klebsiella* spp. including *Klebsiella pneumoniae**, *Proteus mirabilis*, *Proteus* spp. not including *P. penneri* and *P. vulgaris*, *Providencia* spp.

Gram-Positive Anaerobes: *Clostridium* spp. not including *C. difficile*.

Gram-Negative Anaerobes: *Bacteroides* spp. not including *B. fragilis*, *Fusobacterium* spp.

Inherently resistant organisms are:

Gram-Positive Aerobes: *Enterococcus* spp. including *E. faecalis* and *E. faecium*, *Listeria monocytogenes*.

Gram-Negative Aerobes: *Acinetobacter* spp., *Burkholderia cepacia*, *Campylobacter* spp., *Citrobacter freundii*, *Enterobacter aerogenes*, *Enterobacter cloacae*, *Morganella morganii*, *Proteus penneri*, *Proteus vulgaris*, *Pseudomonas* spp. including *Pseudomonas aeruginosa*, *Serratia* spp., *Stenotrophomonas maltophilia*.

Gram-Positive Anaerobes: *Clostridium difficile*.

Gram-Negative Anaerobes: *Bacteroides fragilis*.

Others: Chlamydia species, *Mycoplasma* species, *Legionella* species.

PHARMACOKINETICS

Absorption

After oral administration, Xylid is slowly absorbed from the gastrointestinal tract and rapidly hydrolysed in the intestinal mucosa and blood to release cefuroxime into the circulation. Optimum absorption occurs when it is administered shortly after a meal.

Following administration of Xylid tablets peak serum levels (2.1 mg/l for a 125 mg dose, 4.1 mg/l for a 250 mg dose, 7.0 mg/l for a 500 mg dose and 13.6 mg/l for a 1 g dose) occur approximately 2 to 3 hours after dosing when taken with food.

Distribution

Protein binding has been variously stated as 33 to 50% depending on the methodology used.

Metabolism

Cefuroxime is not metabolised.

Elimination

The serum half-life is between 1 and 1.5 hours.

Cefuroxime is excreted by glomerular filtration and tubular secretion. Concurrent administration of probenecid increases the area under the mean serum concentrations time curve by 50%.

Renal impairment:

Cefuroxime pharmacokinetics have been investigated in patients with various degrees of renal impairment. Based on data from published literature, cefuroxime elimination half-life

increases with decrease in renal function which serves as the basis for dosage adjustment recommendations in this group of patients (See Dosage and Administration). Findings from published data reports that in patients undergoing haemodialysis, at least 60% of the total amount of cefuroxime present in the body at the start of dialysis will be removed during a 4-hour dialysis period. Therefore, an additional single dose of cefuroxime should be administered following the completion of haemodialysis.

INDICATIONS

Xylid is an oral prodrug of the bactericidal cephalosporin antibiotic cefuroxime, which is resistant to most β (beta)-lactamases and is active against a wide range of Gram-positive and Gram-negative organisms.

It is indicated for the treatment of infections caused by susceptible bacteria. Susceptibility to Xylid will vary with geography and time and local susceptibility data should be consulted where available.

Indications include:

- Upper respiratory tract infections for example, ear, nose and throat infections, such as otitis media, sinusitis, tonsillitis and pharyngitis
- Lower respiratory tract infections for example, pneumonia, acute bronchitis, and acute exacerbations of chronic bronchitis
- Genito-urinary tract infections for example, pyelonephritis, cystitis and urethritis
- Skin and soft tissue infections for example, furunculosis, pyoderma and impetigo
- Gonorrhoea, acute uncomplicated gonococcal urethritis, and cervicitis

CONTRAINDICATIONS

Xylid is contraindicated in patients with known hypersensitivity to cephalosporin antibiotics.

WARNING AND PRECAUTIONS

Hypersensitivity

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Xylid, careful inquiry should be made concerning previous hypersensitivity reaction to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Xylid must be discontinued immediately and appropriate alternative therapy instituted.

Overgrowth of non-susceptible organisms

As with other antibiotics, use of Xylid may result in the overgrowth of *Candida*. Prolonged use may also result in the overgrowth of other non-susceptible organisms (e.g. enterococci and *Clostridium difficile*), which may require interruption of treatment.

Pseudomembranous colitis

Pseudomembranous colitis has been reported with the use of antibiotics, and may range in severity from mild to life-threatening. Therefore, it is important to consider its diagnosis in patients who develop diarrhoea during or after antibiotic use. If prolonged or significant diarrhoea occurs or the patient experiences abdominal cramps, treatment should be discontinued immediately and the patients investigated further. When the colitis is not relieved by drug discontinuation or when it is severe, oral vancomycin is the treatment of choice for antibiotic-associated pseudomembranous colitis produced by *Clostridium difficile*. Other causes of colitis should be also considered.

Kidney function

Although Xylid rarely produces alterations in kidney function, evaluation of renal status during therapy is recommended, especially in seriously ill patients receiving the maximum doses.

The total daily doses of Xylid should be reduced in patients with transient or persistent renal insufficiency because high and prolonged serum antibiotic concentrations can occur in such individuals from usual doses.

Nephrotoxicity has been reported following concomitant administration of aminoglycoside antibiotics and cephalosporins.

Therapy

With a sequential therapy regimen, the timing of change to oral therapy is determined by severity of the infection, clinical status of the patient and susceptibility of the pathogens involved. If there is no clinical improvement within 72 hours, the parenteral course of treatment must be continued.

Effects on the Ability to Drive or Operate Machinery

As this medicine may cause dizziness, patients should be warned to be cautious when driving or operating a machinery.

Pregnancy

There is no experimental evidence of embryopathic or teratogenic effects attributable to Xylid but, as with all drugs, it should be

administered with caution during the early months of pregnancy.

Lactation

Since cefuroxime is excreted in human milk, caution should be exercised when Xylid is administered to a nursing mother.

Paediatric Use

There is no experience of using Xylid in children less than 3 months.

DRUG INTERACTION

Drugs which reduce gastric acidity may result in a lower bioavailability of Xylid compared with that of the fasting state and tend to cancel the effect of enhanced postprandial absorption.

In common with other antibiotics, Xylid may affect the gut flora, leading to lower oestrogen reabsorption and reduced efficacy of combined oral contraceptives.

As a false negative result may occur in the ferricyanide test, it is recommended that either the glucose oxidase or hexokinase methods are used to determine blood/plasma glucose levels in patients receiving Xylid. This antibiotic does not interfere in the alkaline picrate assay for creatinine.

ADVERSE REACTIONS

Adverse drug reactions to Xylid are generally mild and transient in nature.

The following convention has been used for the classification of frequency:
very common ≥1/10
common ≥1/100 and <1/10
uncommon ≥1/1000 and <1/100
rare ≥1/10,000 and <1/1000
very rare <1/10,000.

Infections and Infestations

Common: Overgrowth of Candida.

Blood and Lymphatic System Disorders

Common: Eosinophilia
Uncommon: Positive Coomb's test, thrombocytopenia and leukopenia (sometimes profound; sore throat can be a sign of lack of white blood cells).
Very rare: Haemolytic anaemia

Cephalosporins as a class tend to be absorbed onto the surface of red cells membranes and react with antibodies directed against the drug to produce a positive Coomb's test (which can

interfere with cross-matching of blood) and very rarely haemolytic anaemia.

Immune System Disorders (hypersensitivity reactions including)

Uncommon: Skin rashes
Rare: Urticaria and pruritus
Very rare: Drug fever, serum sickness and Anaphylaxis

Nervous System Disorders

Common: Headache and dizziness Gastrointestinal Disorders
Common: Gastrointestinal disturbances including diarrhoea, nausea and abdominal pain
Uncommon: Vomiting
Rare: Pseudomembranous colitis

Hepatobiliary Disorders

Common: Transient increase of hepatic enzyme levels, [alanine transaminase (ALT) serum glutamic-pyruvic transaminase (SGPT); aspartate transaminase (AST) serum glutamic-oxaloacetic transaminase (SGOT), lactate dehydrogenase (LDH)]
Very rare: Jaundice (predominantly cholestatic and hepatitis

Skin and Subcutaneous Tissue Disorders

Very rare: Erythema multiforme, Steven-Johnson syndrome and toxic epidermal necrolysis (exanthematic necrolysis) (see Immune System Disorders as previously mentioned).

DOSAGE AND ADMINISTRATION

The usual course of therapy is 7 days (range 5 - 10 days). Xylid tablets should be taken after food for optimum absorption.

Adults:

Most infections	- 250 mg twice daily
Urinary tract infections	- 250 mg twice daily
Mild to moderate lower tract infections e.g. bronchitis	- 250 mg twice daily
More severe lower tract infections, or if pneumonia is suspected	- 500 mg twice daily
Pyelonephritis	- 250 mg twice daily
Uncomplicated gonorrhoea	- single dose of 1 g

Sequential therapy:

Pneumonia:
1.5 g cefuroxime sodium twice daily (given i.v. or i.m.) for 48 to 72 hours, followed by 500 mg twice daily Xylid tablet (cefuroxime axetil) oral therapy for 7 to 10 days.

Acute exacerbations of chronic bronchitis:
750 mg cefuroxime sodium twice daily (given i.v. or i.m.) for 48 to 72 hours, followed by 500 mg twice daily Xylid tablet (cefuroxime axetil) oral therapy for 5 to 10 days.

Duration of both parenteral and oral therapy is determined by the severity of the infection and the clinical status of the patient.

Children:

Most infections 125 mg (1 x 125 mg tablet) twice daily.

Children with otitis media or where appropriate, with more severe infections, 250 mg (1 x 250 mg tablet or 2 x 125 mg tablets) twice daily.

Xylid tablets should not be crushed or split and are therefore unsuitable for treatment of patients, such as younger children, who cannot swallow whole tablets.

There is no experience of using Xylid tablet in children under the age of 3 months.

Renal impairment:

Cefuroxime is primarily excreted by kidneys. In patients with markedly impaired renal function it is recommended that the dosage of cefuroxime be reduced to compensate for its slower excretion.

For creatinine clearance of ≥30 ml/min (T 1/2 = 1.4 - 2.4 hours), no dose adjustment necessary standard dose of 125 mg to 500 mg given twice daily.

For creatinine clearance of 10-29 ml/min (T 1/2 = 4.6 hours), standard individual dose given every 24 hours.

For creatinine clearance of <10 ml/min (T 1/2 = 16.8 hours), standard individual dose given every 48 hours.

During haemodialysis (T 1/2 = 2 – 4 hours), a single additional standard individual dose should be given at the end of each dialysis.

OVERDOSAGE

Overdosage of cephalosporins can cause cerebral irritation leading to convulsions. Serum level of Xylid can be reduced haemodialysis and peritoneal dialysis.

ROUTE OF ADMINISTRATION

Oral

STORAGE CONDITION

Store below 30°C.
Protect from light.

SHELF - LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

Pack of 10, 50 and 100 tablets in blisters.

Date of revision: 12th June 2024

PRODUCT REGISTRATION HOLDER / MANUFACTURER:

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